

EE214 Lab-3 Report

Name: Medhansh

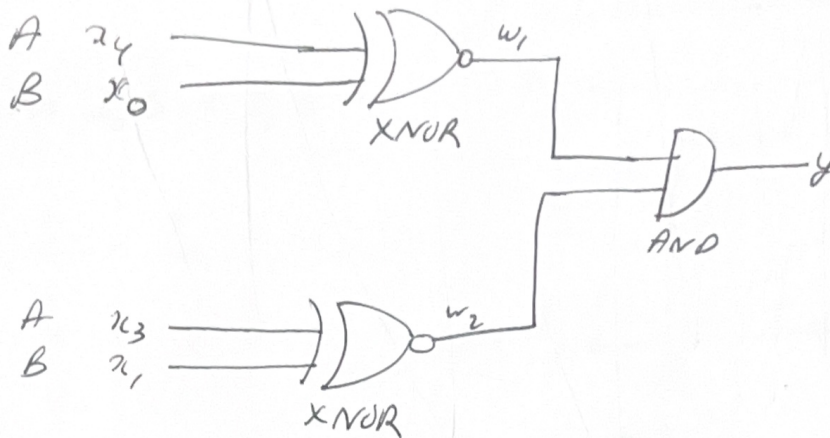
Roll No.: 25B3986

EE214-Lab-3

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Palindrome Detector



x_4	x_3	x_1	x_0	y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

$x_2 \Rightarrow$ doesn't matter

$x_4 x_3$ \ $x_1 x_0$	00	01	11	10
00	1	0	0	0
01	0	0	0	1
11	0	0	1	0
10	0	1	0	0

$$y = \bar{x}_0 \bar{x}_1 \bar{x}_3 \bar{x}_4 + \bar{x}_0 \bar{x}_1 x_3 \bar{x}_4 + \bar{x}_0 x_1 x_3 \bar{x}_4 + x_0 x_1 x_3 x_4$$

Simplified:

$$y = (x_4 \oplus x_0) \cdot (x_3 \oplus x_1)$$

Verified
21/08/21
@Kunalp

Palindrome Detector

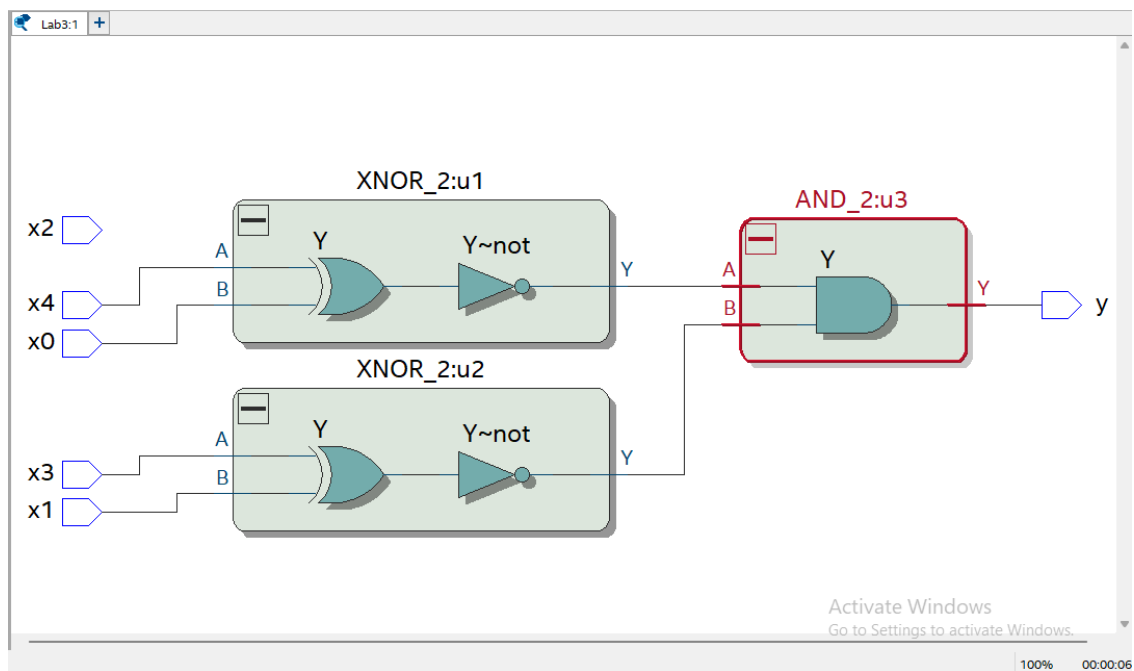


Figure 1: Schematic

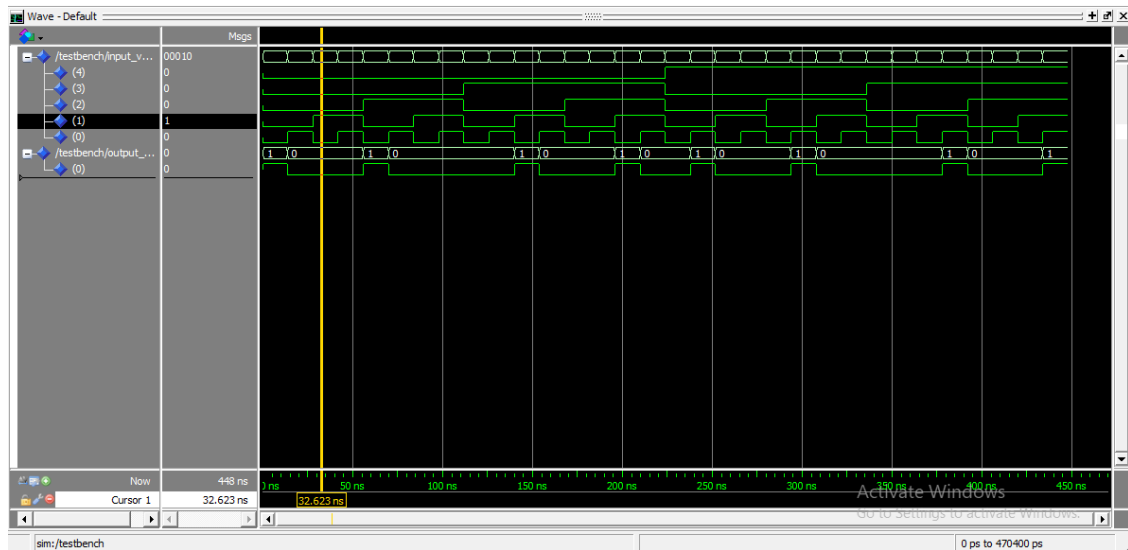


Figure 2: Waveform

```
#
# add wave *
# view structure
# .main_pane.structure.interior.cs.body.struct
# view signals
# .main_pane.objects.interior.cs.body.tree
# run -all
# ** Note: SUCCESS, all tests passed.
#   Time: 448 ns   Iteration: 0   Instance: /testbench

/SIM 2> ]
```

Figure 3: Transcript